

Add the following components:

- 1x ESP32
- 4x LEDs (change colors in their properties to Green, Yellow, Red, Red)
- 4x Resistors (set their values to **180 ohm**)
- 1x HC-SR04 Ultrasonic Sensor
- 1x I2C LCD 16x2
- 1x Relay Module
- 1x Motor (to simulate your AC/DC water pump)

2. Wiring Connections Guide

Based on the schematic in your image, here is exactly how you need to route your virtual wires in Wokwi. Click on a pin on one component, and click the destination pin to create a wire.

Component	Component Pin	Connects to ESP32 / Power Pin
Green LED	Anode (via resistor)	Pin D19
Yellow LED	Anode (via resistor)	Pin D18
Red LED 1	Anode (via resistor)	Pin D5
Red LED 2	Anode (via resistor)	Pin D15
All LEDs	Cathode (short leg)	GND
HC-SR04 Sensor	VCC	VIN
HC-SR04 Sensor	Trig	Pin D13
HC-SR04 Sensor	Echo	Pin D12
HC-SR04 Sensor	GND	GND
I2C LCD	VCC	VIN
I2C LCD	SDA	Pin D21
I2C LCD	SCL	Pin D22

I2C LCD	GND	GND
Relay Module	VCC	VIN
Relay Module	IN	Pin D4
Relay Module	GND	GND

Note on the External Power and Pump: In a real-world setup (like your diagram), the relay acts as a switch for an isolated power supply running the water pump. In Wokwi, you can wire a basic DC motor through the relay's "Normally Open" (NO) and "Common" (COM) terminals to an external battery component to simulate the pump turning on and off.